UNIT - 1

Single Degree Freedom Systems

Subject: Mechanical Vibrations | Code: 23A03702a

Syllabus: Un-damped free vibration: classical method, energy method, equivalent systems, torsional systems. Damped free vibration: viscous (under, critical, over damping), Coulomb damping, equivalent damping coefficient. Whirling of shafts: Dunkerley's lower bound, critical speed.

1. Basic Concepts of Vibration

Vibration = repetitive to-and-fro motion of a body (or system) about its mean / equilibrium position.

- *Period (T) : time for one complete cycle, seconds. Frequency (f) = 1/T (Hz).*
- *Circular frequency (ω) = 2πf = 2π/T rad/s. Amplitude (X) = max. displacement from mean position.*
- *Natural frequency (ω_n) : frequency at which a system vibrates by itself after an initial disturbance, with no external force.*
- *Degrees of freedom (DOF) : minimum number of independent co-ordinates required to completely define the position of the system at any instant.*
  - *→ Single DOF → one co-ordinate (x or θ). Two DOF → two co-ordinates, and so on.*
- *Simple Harmonic Motion (SHM) : x = X sin ωt ; velocity = ωX cos ωt ; acceleration = $-\omega^2 x$ (acceleration ~ displacement, directed to mean position).*

Classification of vibration

Basis	Types
External force	**Free** (no force after the initial disturbance) / **Forced** (periodic force acts continuously)
Damping	**Undamped** (no energy loss) / **Damped** (energy dissipated by friction etc.)
Governing equation	**Linear** (superposition holds) / **Non-linear**
Nature of excitation	**Deterministic** (predictable) / **Random** (only statistical description)

Motion	**Longitudinal, Transverse, Torsional**

Elements of a vibrating system



Spring: F = k x
k = stiffness (N/m)

Damper: F = c ẋ
c = damping coeff. (N·s/m)

Mass: F = m ẍ
inertia element (kg)

2. Un-damped Free Vibration of SDOF System

A mass m attached to a spring of stiffness k, free to move along one direction (x).

Disturb it and release - it vibrates at its natural frequency.

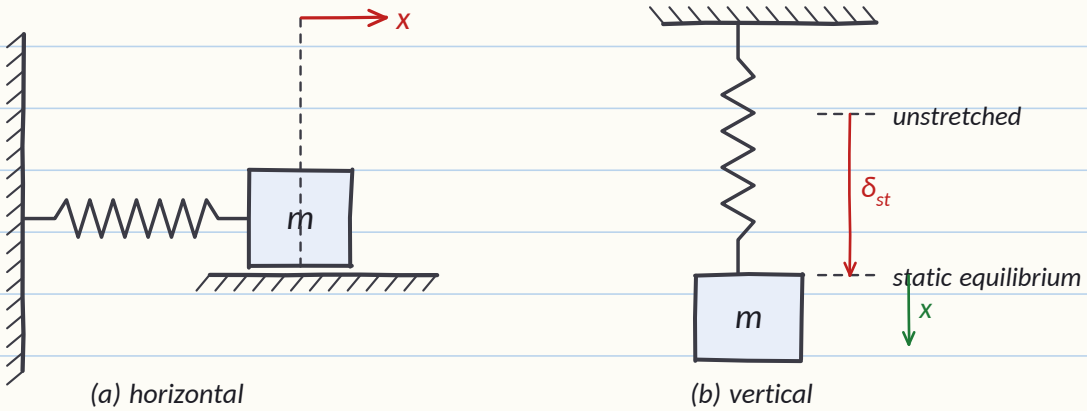


Fig 1.1 Spring-mass system

(a) Classical (Newton's / D'Alembert's) method

Take displacement x from static equilibrium position. In vertical system, spring force at equilibrium balances weight: $k\,\delta_{st} = m\,g$ (so gravity drops out of the equation).

Newton's 2nd law: $m\,\ddot{x} = -k\,x$

$$m\,\ddot{x} + k\,x = 0 \text{ or } \ddot{x} + \omega_n^2\,x = 0 \quad (1)$$

Try $x = e^{st} \rightarrow m\,s^2 + k = 0 \rightarrow s = \pm i\,\sqrt{(k/m)} = \pm i\,\omega_n$

$$\omega_n = \sqrt{(k/m)} \text{ rad/s} = \sqrt{(g/\delta_{st})}$$
$$f_n = (1/2\pi)\sqrt{(k/m)} \text{ Hz } T = 2\pi\sqrt{(m/k)} \text{ s} \quad (2)$$

General solution: $x(t) = A\cos\omega_n t + B\sin\omega_n t$

With initial conditions $x(0) = x_0$, $\dot{x}(0) = \dot{x}_0$:

$$x(t) = x_0\cos\omega_n t + (\dot{x}_0/\omega_n)\sin\omega_n t = X\cos(\omega_n t - \varphi)$$
$$X = \sqrt{(x_0^2 + (\dot{x}_0/\omega_n)^2)}\ \varphi = \tan^{-1}[\,\dot{x}_0 / (x_0\,\omega_n)\,] \quad (3)$$

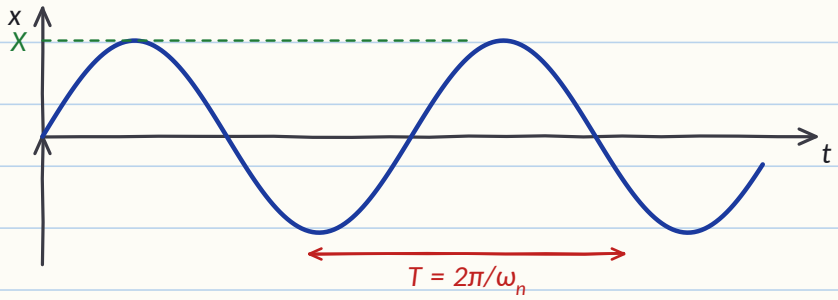


Fig 1.2 Un-damped free response (simple harmonic)

Remember:

Natural frequency depends ONLY on k and m (system properties), not on the initial conditions. Initial conditions decide only amplitude X and phase φ.

(b) Energy method (Rayleigh's method)

For a conservative system total energy is constant: $T + U = \text{constant}$ (T = kinetic, U = potential)

$$d/dt\,(T + U) = 0 \rightarrow \text{equation of motion}$$
$$\text{or } T_{max} = U_{max} \rightarrow \omega_n \quad (4)$$

- For SHM $T_{max} = \tfrac{1}{2}\, m_{eff}\, (\omega_n X)^2$, $U_{max} = \tfrac{1}{2}\, k_{eff}\, X^2 \rightarrow \omega_n = \sqrt{(k_{eff}/m_{eff})}$.
- Steps: (1) choose co-ordinate, (2) write T and U in terms of it, (3) differentiate or equate maxima.

Spring-mass: $T = \tfrac{1}{2}\, m\, \dot{x}^2$, $U = \tfrac{1}{2}\, k\, x^2$. $d/dt(T+U) = (m\,\ddot{x} + k\,x)\,\dot{x} = 0 \rightarrow m\,\ddot{x} + kx = 0$ (same as Newton).

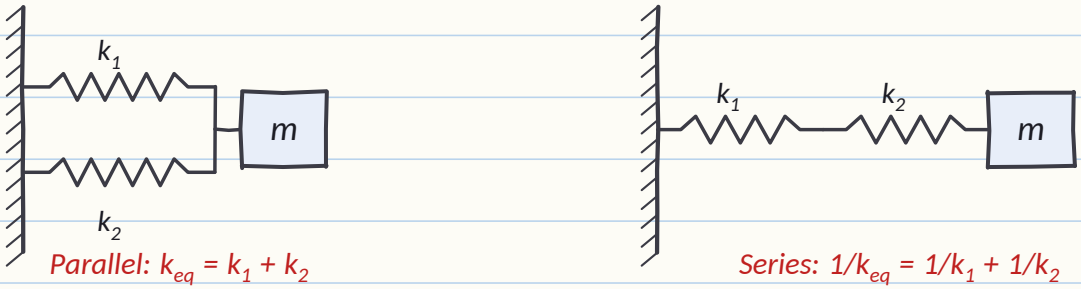
Effect of spring mass (m_s): a small element dy of spring moves with velocity $(y/l)\dot{x}$.

$T_{spring} = \int \tfrac{1}{2}\, (m_s/l)\, dy\, (y\,\dot{x}/l)^2 = \tfrac{1}{6}\, m_s\, \dot{x}^2 \rightarrow$ effective mass $= m + m_s/3$

$$\omega_n = \sqrt{(\, k \,/\, (m + m_s/3)\,)} \quad (5)$$

3. Equivalent Systems (Equivalent Stiffness)

Several springs can be replaced by one equivalent spring k_{eq} without changing the natural frequency.



Parallel: $k_{eq} = k_1 + k_2$ Series: $1/k_{eq} = 1/k_1 + 1/k_2$

Fig 1.3 Spring combinations

Elastic member (end mass)	Equivalent stiffness k
Cantilever, load at free end (length l)	$3EI / l^3$
Simply supported beam, load at centre	$48EI / l^3$
Fixed-fixed beam, load at centre	$192EI / l^3$
Axial rod (area A, length l)	AE / l
Torsion of shaft (length l)	$k_t = GJ / l$ (N·m/rad)

Beams in series with spring: add flexibilities (1/k). Beam and spring in parallel: add stiffnesses.

4. Torsional Vibration Systems

A disc of mass moment of inertia I ($kg·m^2$) on a shaft of length l, diameter d.

Angular displacement θ about the shaft axis.

Torsional stiffness $k_t = T/\theta = GJ/l$, polar moment $J = \pi d^4/32$.

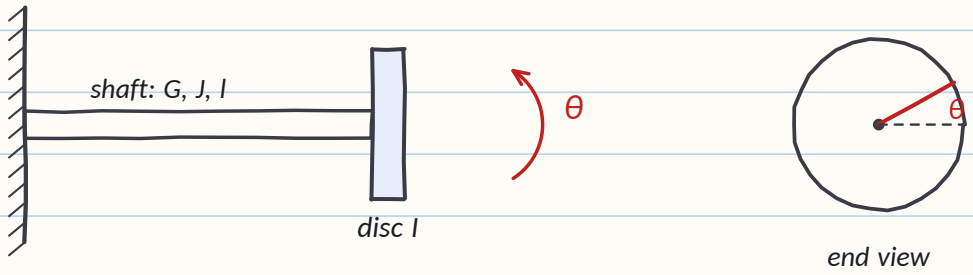


Fig 1.4 Torsional pendulum

Equation of motion: $I\ \ddot{\theta} + k_t\ \theta = 0$ ($\ddot{\theta}$ = angular acceleration)

$$\omega_n = \sqrt{(k_t/I)} = \sqrt{(\ GJ\ /\ (I\ l)\ )}\ rad/s \quad (6)$$

Two-rotor system

Two discs I_1, I_2 on a shaft, with torsional stiffness $k_t = GJ/l$ (between the discs). They vibrate in opposite directions; a node (zero amplitude section) lies between them.

Node position (from disc 1): both sides must have the same frequency, $\omega_n^2 =$

$GJ/(l_1 l_1) = GJ/(l_2 l_2)$, with $l_1 + l_2 = l$.

$$l_1 = l\, I_2 / (I_1 + I_2) \quad l_2 = l\, I_1 / (I_1 + I_2)$$
$$\omega_n = \sqrt{(GJ\, (I_1 + I_2) / (I_1\, I_2\, l))} \qquad (7)$$

- Geared system: refer all inertias/stiffnesses to one shaft using gear ratio $n = N_2/N_1$. $I_{equiv} = n^2 I$ (referred to motor shaft); $k_{t,equiv} = n^2 k_t$.

5. Damped Free Vibration

Real systems lose energy (friction, air, internal). Damping force opposes velocity.

Types: viscous, Coulomb (dry friction), hysteretic (structural).

5.1 Viscous damping ($F_d = c\,\dot{x}$)

Damper force proportional to velocity. Newton: $m\,\ddot{x} = - k\,x - c\,\dot{x}$

$$m\,\ddot{x} + c\,\dot{x} + k\,x = 0 \rightarrow \ddot{x} + 2\zeta\omega_n\,\dot{x} + \omega_n^2\,x = 0 \qquad (8)$$

Critical damping coefficient $c_c = 2\sqrt{(km)} = 2m\omega_n$ (dividing line between oscillatory & non-oscillatory motion). Damping ratio $\zeta = c/c_c$.

Characteristic eqn ($x = e^{st}$): $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$

$$s_{1,2} = - \zeta\omega_n \pm \omega_n \sqrt{(\zeta^2 - 1)} \qquad (9)$$

under damped ($\zeta < 1$)
critically damped ($\zeta = 1$)
over damped ($\zeta > 1$)

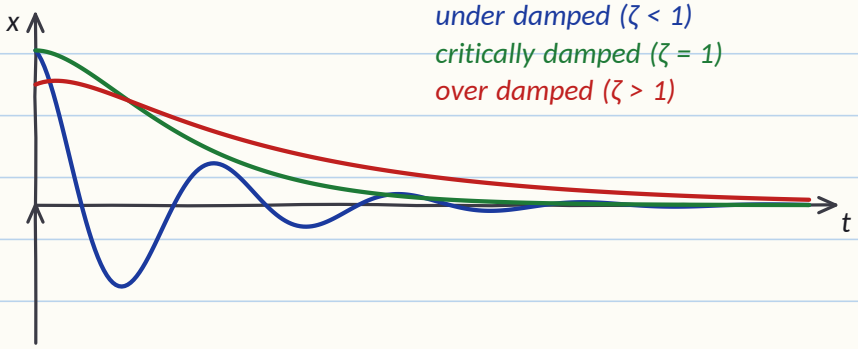


Fig 1.5 Response of viscously damped system

Case 1 : Under-damped ($\zeta < 1$, $c < c_c$)

Roots complex: $s = -\zeta\omega_n \pm i\,\omega_d$, $\omega_d = \omega_n\sqrt{(1 - \zeta^2)}$ (damped natural frequency)

$$x(t) = e^{-\zeta\omega_n t} (A \cos \omega_d t + B \sin \omega_d t) = X\, e^{-\zeta\omega_n t} \cos(\omega_d t - \varphi)$$
$$A = x_0,\ B = (\dot{x}_0 + \zeta\omega_n x_0) / \omega_d \qquad (10)$$

- Oscillatory, amplitude decays exponentially (envelope $X\, e^{-\zeta\omega_n t}$). Time period $T_d = 2\pi/\omega_d$ (slightly more than undamped).

Case 2 : Critically damped ($\zeta = 1$, $c = c_c$)

Equal real roots $s = -\omega_n$. $x(t) = (A + B\,t)\,e^{-\omega_n t}$, $A = x_0$, $B = \dot{x}_0 + \omega_n x_0$

- No oscillation. Returns to equilibrium in minimum time. Used in guns recoil mechanism, door closers, instruments.

Case 3 : Over-damped ($\zeta > 1$, $c > c_c$)

Two distinct negative real roots $s_{1,2} = (-\zeta \pm \sqrt{(\zeta^2 - 1)})\,\omega_n$.

$x(t) = A\,e^{s_1 t} + B\,e^{s_2 t}$ - non-oscillatory, aperiodic, slow return to equilibrium.

Logarithmic decrement (δ)

Natural log of the ratio of any two successive amplitudes (one period T_d apart) of an under-damped system:

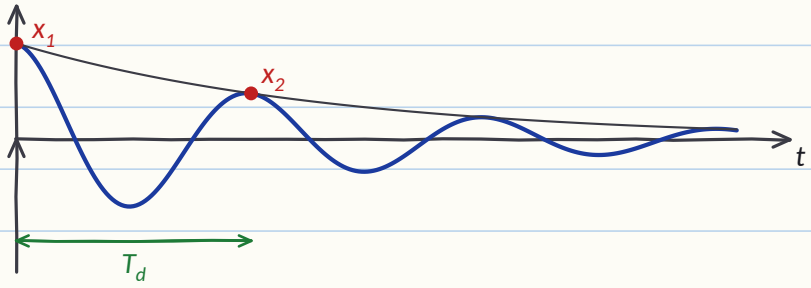


Fig 1.6 Successive amplitudes

$$\delta = \ln(x_1/x_2) = 2\pi\zeta / \sqrt{(1 - \zeta^2)} \approx 2\pi\zeta \text{ (for small } \zeta)$$

$$\text{after n cycles: } \delta = (1/n)\ln(x_0/x_n)\quad \zeta = \delta / \sqrt{(4\pi^2 + \delta^2)} \qquad (11)$$

Use:

From experiment: measure x_0 and x_n, find δ, then ζ and then $c = \zeta\, c_c$. This is the standard way of finding damping of a system.

5.2 Coulomb (dry-friction) damping

Friction force $F = \mu N = \mu mg$ (constant magnitude) always opposes the direction of motion. Mass sliding on rough surface with spring.

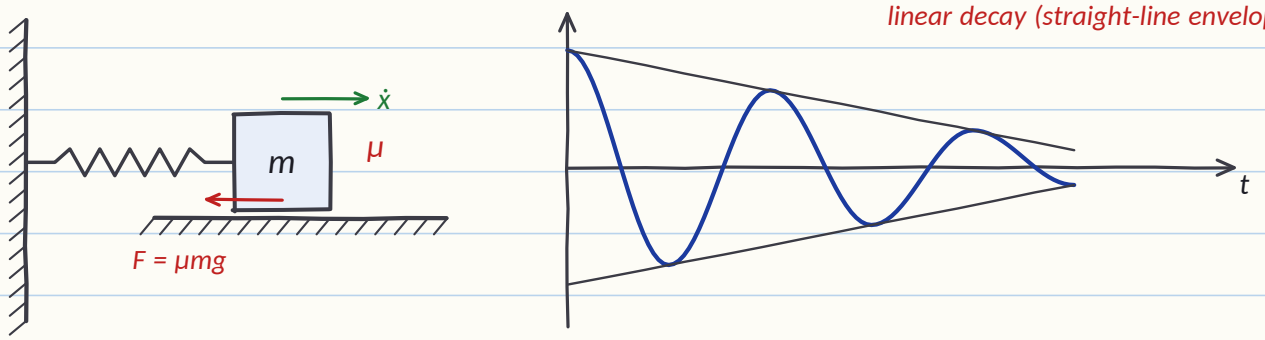


Fig 1.7 Coulomb damped system and response

- Equation: $m\ddot{x} + kx = -F$ when $\dot{x} > 0$ (moving right); $m\ddot{x} + kx = +F$ when $\dot{x} < 0$. Solution: $x = A\cos\omega_n t + B\sin\omega_n t \pm F/k$.
- Frequency is same as undamped: $\omega_n = \sqrt{(k/m)}$. The equilibrium position shifts to $\pm F/k$ alternately in each half cycle.
- Amplitude reduces by $2F/k$ per half cycle and $4F/k$ per full cycle (linear decay, not exponential).
- Motion stops when amplitude $\leq F/k$ (spring force cannot overcome friction). Final rest position may not be at $x = 0$.

$$x_n = x_0 - n\,(4F/k) \quad \text{No. of cycles} = (x_0 - F/k) / (4F/k)\ ,\ F = \mu m g \qquad (12)$$

5.3 Equivalent viscous damping coefficient (c_{eq})

To simplify, replace any non-viscous damping by an equivalent viscous damper that dissipates same energy per cycle at harmonic motion $x = X\sin\omega t$.

Viscous damper: $\Delta W = \int F_d\,dx = \int (c\dot{x})\dot{x}\,dt = \pi c \omega X^2$

Coulomb: $\Delta W = 4FX$ (force F over distance $4X$ per cycle). Equate: $\pi c_{eq}\omega X^2 = 4FX$

$$\text{Coulomb: } c_{eq} = 4F / (\pi\omega X)$$
$$\text{Quadratic } (F = c'\dot{x}^2)\text{: } c_{eq} = 8c'\omega X / (3\pi) \qquad (13)$$
$$\text{Hysteretic } (\Delta W = \pi h X^2)\text{: } c_{eq} = h/\omega$$

- Note: for Coulomb damping c_{eq} depends on X and ω - so it is amplitude dependent.

6. Solved Problems

Problem 1 (natural frequency from static deflection)

A mass of 10 kg suspended by a spring deflects 2 mm statically. Find k, ω_n, f_n.

$k = mg/\delta_{st} = 10(9.81)/0.002 = 49\,050$ N/m

$\omega_n = \sqrt{(g/\delta_{st})} = \sqrt{(9.81/0.002)} = 70.04$ rad/s; $f_n = \omega_n/2\pi = 11.15$ Hz

Problem 2 (energy method - uniform rod)

A uniform rod, mass m, length l, pivoted at one end swings in vertical plane. Find ω_n for small oscillations.

$I_O = ml^2/3$. $T = \tfrac{1}{2} I_O \dot{\theta}^2 = \tfrac{1}{6} m l^2 \dot{\theta}^2$. $U = mg(l/2)(1 - \cos\theta) \approx \tfrac{1}{4} mgl\theta^2$

$T_{max} = U_{max}$: $\tfrac{1}{2}(ml^2/3)\omega_n^2 = \tfrac{1}{2}(mgl/2) \rightarrow \omega_n = \sqrt{(3g/2l)}$.

For $l = 1$ m: $\omega_n = \sqrt{14.715} = 3.836$ rad/s, $T = 2\pi/\omega_n = 1.64$ s.

Problem 3 (cantilever with end mass)

Steel cantilever $E = 200$ GPa, $l = 0.4$ m, section 20 mm × 10 mm (thickness 10 mm) carries 5 kg at the free end. Find f_n. Neglect beam mass.

$I = b\,h^3/12 = 0.02(0.01)^3/12 = 1.667\times10^{-9}\ m^4$

$k = 3EI/l^3 = 3(200\times10^9)(1.667\times10^{-9})/(0.4)^3 =$ 15 625 N/m

$\omega_n = \sqrt{(15625/5)} = 55.9$ rad/s; $f_n =$ 8.90 Hz

Problem 4 (torsional system)

A disc of $I = 0.1$ kg·m^2 is fixed to a steel shaft ($G = 80$ GPa), $d = 20$ mm, $l = 0.5$ m, other end fixed. Find f_n.

$J = \pi d^4/32 = 1.571\times10^{-8}\ m^4$; $k_t = GJ/l = 80\times10^9(1.571\times10^{-8})/0.5 = 2513$ N·m/rad

$\omega_n = \sqrt{(k_t/I)} = \sqrt{(2513/0.1)} = 158.5$ rad/s; $f_n =$ 25.2 Hz

Problem 5 (viscous damping)

$m = 10$ kg, $k = 4000$ N/m, $c = 40$ N·s/m. Find ζ, ω_d, δ and ratio of successive amplitudes.

$\omega_n = \sqrt{(4000/10)} = 20$ rad/s; $c_c = 2\sqrt{(km)} = 2\sqrt{(40000)} = 400$ N·s/m; $\zeta = 40/400 =$ 0.1 (under-damped)

$\omega_d = 20\sqrt{(1 - 0.01)} =$ 19.90 rad/s; $\delta = 2\pi(0.1)/\sqrt{0.99} =$ 0.6315

$x_1/x_2 = e^{\delta} = e^{0.6315} =$ 1.88 (each amplitude is about 53% of the previous one).

Problem 6 (finding ζ from decay)

Vibration amplitude of a system falls to 25% of initial value after 5 cycles. Find ζ.

$\delta = (1/5)\ln(1/0.25) = 0.2\ (1.386) = 0.2773$

$\zeta = \delta / \sqrt{(4\pi^2 + \delta^2)} = 0.2773 / \sqrt{(39.48 + 0.077)} =$ 0.0441 (≈ 4.4 % of critical).

Problem 7 (Coulomb damping)

$m = 2$ kg, $k = 800$ N/m, $\mu = 0.15$. Mass pulled 50 mm and released. Find (i) amplitude loss per cycle, (ii) when the mass stops, (iii) c_{eq} for first cycle.

$F = \mu mg = 0.15(2)(9.81) = 2.943$ N; $F/k = 3.68$ mm; $\omega_n = \sqrt{(800/2)} = 20$ rad/s

(i) Loss per cycle $= 4F/k =$ 14.72 mm (7.36 mm per half cycle).

(ii) Amplitudes after half cycles: 42.6, 35.3, 27.9, 20.6, 13.2, 5.85 (> 3.68, still moves), then -1.51 mm (< F/k) → stops after 7 half cycles (3.5 cycles), 1.51 mm from the mean position. Time $= 7\pi/\omega_n = 1.10$ s.

(iii) $c_{eq} = 4F/(\pi\ \omega\ X) = 4(2.943)/(\pi \times 20 \times 0.05) =$ 3.75 N·s/m.

Problem 9 (equivalent stiffness)

Springs $k_1 = 3000$ and $k_2 = 6000$ N/m are in series; this combination is in parallel with $k_3 = 2000$ N/m. A 10 kg mass is attached. Find f_n.

Series: $k_{12} = k_1k_2/(k_1+k_2) = 18\times10^6/9000 = 2000$ N/m. Parallel: $k_{eq} = 2000 + 2000 =$ 4000 N/m

$\omega_n = \sqrt{(4000/10)} = 20$ rad/s; $f_n = 3.18$ Hz.

Problem 10 (U-tube manometer - energy method)

Liquid column of total length L in a U-tube of area A, density ρ. Find ω_n.

Displace the level by x: mass $m = \rho AL$; $T = \tfrac{1}{2}\,\rho AL\,\dot{x}^2$. Net head difference 2x so $U = \tfrac{1}{2}\,(2\rho gA)\,x^2$.

$T_{max} = U_{max} \rightarrow \omega_n^2 = 2\rho gA / \rho AL \rightarrow \omega_n = \sqrt{(2g/L)}$. For L = 0.5 m: $\omega_n = 6.26$ rad/s, $f_n =$ 1.0 Hz.

Problem 11 (two-rotor torsional system)

Two rotors $I_1 = 0.8$ and $I_2 = 0.4$ kg·m² on a steel shaft d = 40 mm, l = 1 m, G = 80 GPa. Find f_n and node position.

$J = \pi(0.04)^4/32 = 2.513\times10^{-7}$ m⁴; GJ = 20 106 N·m²

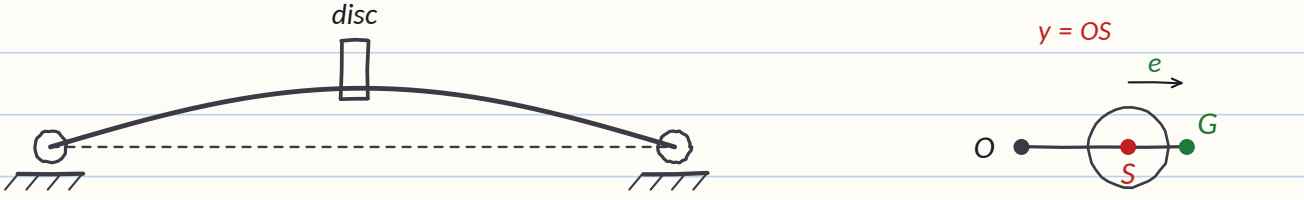
$\omega_n = \sqrt{(\ GJ\ (I_1+I_2)\ /\ (I_1I_2l)\ )} = \sqrt{(20106\times1.2/0.32)} = 274.6$ rad/s, $f_n = 43.7$ Hz

Node: $l_1 = l\ I_2/(I_1+I_2) = 0.4/1.2 = 0.333$ m from the 0.8 kg·m² rotor ($I_2 = 0.667$ m).

7. Whirling of Shafts

Whirling (whipping) : at certain speeds a rotating shaft bows and the bent shaft itself rotates (whirls) in a plane about the bearing axis. Causes: mass eccentricity, non-uniform shaft, unbalance, gyroscopic effects, bearing misalignment.

Critical speed : speed at which whirl amplitude becomes dangerously large (resonance). It equals the natural frequency of transverse vibration of the shaft.



bearings & deflected shaft — O: bearing axis, S: shaft centre, G: C.G.

Fig 1.8 Whirling of a shaft with a disc

Critical speed - derivation (no damping)

Disc mass m, eccentricity e (SG), shaft stiffness k (transverse), shaft deflection y (OS), shaft speed ω.

- Spring (elastic) force on disc, towards O: $k\,y$.
- Centrifugal force on disc (C.G. at distance $y + e$ from O): $m\,\omega^2\,(y + e)$.

Balance: $k\,y = m\,\omega^2\,(y + e) \rightarrow y\,(k - m\omega^2) = m\,\omega^2\,e$

$$y = e\,r^2 / (1 - r^2),\ r = \omega / \omega_n,\ \omega_n = \sqrt{(k/m)} \quad (14)$$

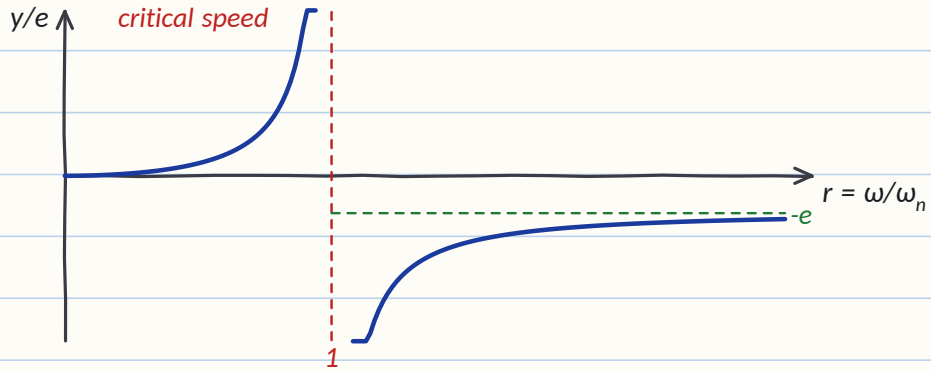


Fig 1.9 Whirl amplitude vs speed ratio

- $r < 1$: y positive; C.G. G lies outside S (heavy side out). y rises to ∞ as $r \rightarrow 1$ (resonance).
- $r = 1$: $\omega = \omega_n$ = critical speed: $\omega_c = \sqrt{(k/m)} = \sqrt{(g/\delta)}$ (δ = static deflection of shaft under disc).
- $r > 1$: y negative, $y \rightarrow -e$ as $r \rightarrow \infty$: G lies between O and S - shaft self-centres; running well above critical speed is smooth (turbines).

$$N_c = (60/2\pi)\,\sqrt{(g/\delta)}\ \text{rpm} \approx 946 / \sqrt{\delta}\ (\delta \text{ in mm}) \quad (15)$$

With damping: $y = e\,r^2 / \sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$ - finite at resonance.

Dunkerley's approximation (lower bound)

Shaft carrying several discs: exact fundamental critical speed is difficult; Dunkerley's empirical formula is simple, gives a lower bound (slightly lower than true value).

$$1/\omega_c^2 = 1/\omega_1^2 + 1/\omega_2^2 + 1/\omega_3^2 + \ldots + 1/\omega_s^2$$
$$\text{with } \omega_i^2 = g / \delta_i\ (\delta_i = \text{deflection under load i acting alone}) \quad (16)$$

- ω_i : critical speed if only the i-th disc acted on the shaft. ω_s : critical speed of shaft alone (its own mass); often ignored if shaft is light.
- Equivalent form: $1/\omega_c^2 = (\delta_1 + \delta_2 + \delta_3 + \ldots)/g$.

Deflection of simply supported shaft (span l) under load W at distance a from one

end, $b = l - a$: $\delta = W a^2 b^2 / (3 E I l)$; at centre $\delta = W l^3 / (48 E I)$.

Problem 8 (Dunkerley)

Shaft $d = 20$ mm, $l = 1$ m, $E = 200$ GPa, simply supported, carries 10 kg at 0.25 m from left bearing and 20 kg at mid-span. Ignoring shaft mass, find the critical speed.

$I = \pi d^4/64 = 7.854\times10^{-9}\ m^4$; $EI = 1570.8\ N\cdot m^2$

$\delta_1 = W_1 a^2 b^2/(3EIl) = 98.1(0.25)^2(0.75)^2/(3\times1570.8\times1) = 0.732$ mm

$\delta_2 = W_2 l^3/(48EI) = 196.2/(48\times1570.8) = 2.602$ mm

$\omega_1^2 = g/\delta_1 = 13\,403$; $\omega_2^2 = g/\delta_2 = 3770$; $1/\omega_c^2 = 7.46\times10^{-5} + 2.65\times10^{-4} = 3.40\times10^{-4}$

$\omega_c = 54.3$ rad/s $\rightarrow N_c = 60\omega_c/2\pi = \approx 518$ rpm.

Problem 12 (critical speed and whirl amplitude)

A 12 mm steel shaft ($E = 200$ GPa), 0.6 m between simple bearings, carries an 8 kg disc at the centre. Find critical speed, and whirl amplitude at 900 rpm if $e = 0.5$ mm.

$I = \pi(0.012)^4/64 = 1.018\times10^{-9}\ m^4$; $\delta = Wl^3/(48EI) =$

$78.48(0.216)/(48\times200\times10^9\times1.018\times10^{-9}) = 1.735$ mm

$\omega_c = \sqrt{(g/\delta)} = \sqrt{(9.81/1.735\times10^{-3})} = 75.2$ rad/s $\rightarrow N_c \approx 718$ rpm

$r = 900/718 = 1.2535$; $y = e\, r^2/(1 - r^2) = 0.5(1.571)/(1 - 1.571) = -1.375$ mm

(magnitude 1.375 mm; negative → above critical, G between O and S).

8. Quick Revision Table

Quantity	Formula
Natural frequency	$\omega_n = \sqrt{(k/m)} = \sqrt{(g/\delta_{st})}$; torsional $\sqrt{(k_t/I)}$
Damping ratio / critical	$\zeta = c/c_c$; $c_c = 2\sqrt{(km)} = 2m\omega_n$
Damped frequency	$\omega_d = \omega_n\sqrt{(1 - \zeta^2)}$
Log decrement	$\delta = \ln(x_1/x_2) = 2\pi\zeta/\sqrt{(1 - \zeta^2)} \approx 2\pi\zeta$
Coulomb decay / cycle	$4F/k$, $F = \mu mg$; stops if amplitude $\le F/k$
Equivalent damping	Coulomb: $4F/(\pi\omega X)$; quadratic: $8c'\omega X/(3\pi)$
Whirl amplitude	$y = e\, r^2/(1 - r^2)$; $\omega_c = \sqrt{(g/\delta)}$
Dunkerley	$1/\omega_c^2 = \Sigma\ 1/\omega_i^2$ (lower bound)

9. Short Questions & Answers

1. **Define degree of freedom.** Minimum number of independent co-ordinates needed to define the configuration of the system.
2. **Why is the natural frequency independent of amplitude?** In a linear system the restoring force is ~ displacement, so $\omega_n = \sqrt{(k/m)}$ depends only on k and m.
3. **What is critical damping?** The minimum damping ($c = 2\sqrt{(km)}$) at which the system does not oscillate; returns to rest in least time.
4. **Define logarithmic decrement.** Natural log of ratio of two successive amplitudes, $\delta = \ln(x_n/x_{n+1}) = 2\pi\zeta/\sqrt{(1-\zeta^2)}$.
5. **How does Coulomb damping differ from viscous?** Force is constant (μN), decay is linear, motion stops in finite time, frequency unchanged; viscous gives exponential decay, changes frequency to ω_d.
6. **What is meant by equivalent viscous damping?** A viscous damper that dissipates the same energy per cycle as the actual (non-viscous) damping.
7. **What is critical speed of a shaft?** Speed at which the rotational speed equals the natural frequency of transverse vibration; whirl amplitude becomes very large.
8. **Why does Dunkerley's method give a lower bound?** It neglects the interaction of loads and adds flexibilities; the result is always slightly below the true fundamental frequency.
9. **Why is it safe to run above critical speed?** For $r > 1$ the shaft self-centres ($y \to -e$); the speed must be passed through quickly to avoid build-up of amplitude.
10. **Effect of spring mass on ω_n?** One third of spring mass is added to the attached mass: $m_{eff} = m + m_s/3$, so ω_n reduces.

Important questions: (1) derive ω_n by energy method (2) three damping cases with sketches (3) log decrement derivation (4) Coulomb damping decay & c_{eq} (5) critical speed of shaft with a disc (6) Dunkerley's method problem.